

Assessment Book: Mathematics

Period 1 Assessment

1. [MULTIPLE_CHOICE] Which of the following best defines a linear equation in one variable?

- A. An equation with two variables of any power
- B. An equation containing only one variable raised to the first power
- C. An expression with no equals sign
- D. A statement with no variable at all

Answer: An equation containing only one variable raised to the first power

2. [SHORT_ANSWER] In the equation $6x - 4 = 20$, name the variable, coefficient, and constants.

Answer: Variable: x ; Coefficient: 6; Constants: -4 and 20

3. [TRUE_FALSE] True or False: ' $2x + 5$ ' is an equation because it contains a variable.

- A. True
- B. False

Answer: False

4. [APPLICATION] A balanced scale shows the equation $x + 3 = 9$. Using the properties of equality, explain step-by-step how you would find the value of x , describing what happens to the balance at each step.

Answer: Subtract 3 from both sides (property of equality) to keep balance: $x + 3 - 3 = 9 - 3$, so $x = 6$. The balance stays level because equal amounts are removed from both sides.

Period 2 Assessment

1. [SHORT_ANSWER] Solve for x : $x + 9 = 4$.

Answer: $x = -5$

2. [MULTIPLE_CHOICE] Which property justifies dividing both sides of $5x = 20$ by 5 to solve for x ?

- A. Property of Equality
- B. Commutative Property
- C. Distributive Property
- D. Associative Property

Answer: Property of Equality

3. [SHORT_ANSWER] Solve and verify: $3x - 5 = 7$. Show the verification step.

Answer: $x = 4$; verification: $3(4) - 5 = 12 - 5 = 7$, $LHS = RHS$

4. [ANALYSIS] A student solves $2x + 6 = 10$ and gets $x = 8$ by only subtracting 6 and forgetting to divide by 2. Identify the error and give the correct solution with verification.

Answer: Error: student didn't divide both sides by 2 after subtracting 6. Correct: $2x = 4$, $x = 2$; verification: $2(2) + 6 = 10$, $LHS = RHS$.

Period 3 Assessment

1. [SHORT_ANSWER] What happens to the sign of a term when it is transposed from one side of an equation to the other?

Answer: The sign of the term reverses (positive becomes negative and negative becomes positive; multiplication becomes division and vice versa).

2. [PROBLEM_SOLVING] Solve for x using transposition: $6x - 9 = 21$. Then verify your solution by substitution.

Answer: $x = 5$; Verification: $6(5) - 9 = 30 - 9 = 21$, which matches the right side.

3. [ANALYSIS] A student solved $3x + 6 = 15$ by writing $3x = 15 + 6$, giving $x = 7$. Identify the mistake and correct the solution.

Answer: The student transposed $+6$ incorrectly (should become -6 on the other side). Correct: $3x = 15 - 6 = 9$, so $x = 3$.

Period 4 Assessment

1. [MCQ] When a term is transposed from one side of an equation to the other, what happens to it?

- A. It stays the same
- B. Its operation/sign reverses
- C. It is deleted
- D. It becomes squared

Answer: Its operation/sign reverses

2. [SHORT ANSWER] Solve the equation: $5x + 4 = 2x + 19$. Show your transposition steps clearly.

Answer: $x = 5$ ($5x - 2x = 19 - 4 \rightarrow 3x = 15 \rightarrow x = 5$)

3. [ERROR ANALYSIS] A student solved $4x + 6 = x + 18$ and wrote: ' $4x + x = 18 - 6$, so $5x = 12$, $x = 2.4$ '. Identify the student's error and provide the correct solution.

Answer: Error: the student added x instead of subtracting it when transposing (x should move as $-x$, not $+x$). Correct: $4x - x = 18 - 6 \rightarrow 3x = 12 \rightarrow x = 4$.

4. [CONCEPTUAL/VERIFICATION] After solving an equation with variables on both sides, why should you substitute your answer back into both the original left-hand side and right-hand side separately, rather than just one side?

Answer: Substituting into both sides separately and checking that they give equal values confirms the solution is correct and helps catch errors such as incorrect sign changes made during transposition.

Period 5 Assessment

1. [SHORT_ANSWER] Translate into an equation: 'Twice a number decreased by 7 is 15.' Then solve for the number.

Answer: $2x - 7 = 15$; $x = 11$

2. [PROBLEM_SOLVING] The sum of three consecutive integers is 72. Find the integers, showing your equation and solution steps.

Answer: $x + (x+1) + (x+2) = 72 \rightarrow 3x+3=72 \rightarrow x=23$; integers are 23, 24, 25

3. [SHORT_ANSWER] Solve for x : $6x + 5 = 2x + 25$. Identify why this equation has variables on both sides.

Answer: $6x+5=2x+25 \rightarrow 4x=20 \rightarrow x=5$; both sides contain terms with x before simplifying.

4. [EVALUATIVE] A word problem states 'the number of students in a class' and your equation gives $x = -4$. Is this solution acceptable? Justify your reasoning using the context of the problem.

Answer: No, because the number of students cannot be negative; the solution is not reasonable given the real-world context.